Online Auction and Bidding System
Project Description

Spring 2026

Contents

1 Introduction

This project focuses on modeling and building an online bidding marketplace similar to eBay. The platform supports real-time auction and bid updates for buyers, sellers, and administrators, with a PostgreSQL backend for efficient data management. The goal is to deliver a functional demo with a clean, user-friendly interface and a reliable database-driven architecture. Sellers can list items, buyers can place bids and monitor auctions, and administrators can manage users, items, payments, and shipments.
The project is divided into three phases:

- *(i)* ER-based requirements analysis
- *(ii)* Relational schema design
- *(iii)* Implementation.

2 Project Phases

Quick links: Requirement Analysis, Grading Policy, Submission Summary.

2.1 Phase 1: ER Design

In Phase 1, we perform requirement analysis using the ER model. The system requirements cover users, sellers, items, auctions, bids, payments, and shipment data.

Deliverables:

- ER diagram using course-standard notation
- Supporting assumptions and documentation
- One compressed submission file uploaded to Canvas.

Evaluation note: Evaluation focuses on ER correctness and completeness. Assumptions are acceptable if they are clearly documented and do not conflict with project requirements.
Due date: April 25.

2.2 Phase 2: Relational Schema Design

In Phase 2, a common class ER diagram is provided, and each group translates it into a PostgreSQL relational schema.

Deliverables:

- One recreatable `.sql` script
- Table definitions, primary keys, foreign keys, and enforceable constraints
- Submission through Canvas

Evaluation note: Evaluation focuses on correctness and completeness of schema translation. Any constraints not representable purely at the relational level must be documented as limitations or assumptions.
Due date: May 5.

2.3 Phase 3: Implementation

In Phase 3, We will provide a common schema that the whole class will use to implement the complete system.

Core tasks include:

- Physical database design and performance tuning (e.g., indexing)
- Client application development
- Complete technical documentation

The system backend requiers the usage of **PostgreSQL**. The client application enables seller, buyer, and administrator workflows. **Python** is preferred for implementation, but other languages may be used if approved. This phase includes system demo scheduling and presentation. Slots are first-come, first-served.

Deliverables: Final report and source code must be submitted via Canvas before the due date. A dataset may be provided and can be modified if all changes are reported in final documentation.

Evaluation note: Evaluation includes implementation completeness, UI quality, code quality, and documentation quality. Extra credit may be awarded for stronger usability, added features, and robust error handling.
Due date: June 8.

3 Grading Policy

Project grading is distributed as follows:

- **Phase 1 (30%):** Conceptual design (ER diagram correctness and completeness).
- **Phase 2 (10%):** Logical design (accurate ER-to-schema translation).
- **Phase 3 (60%):**
  - 30% SQL queries/reports and core functionality;
  - 10% physical database design and performance tuning;
  - 10% client application development;
  - 10% documentation quality.
- **Extra credit (+10%):** user-friendly GUI, additional features, or meaningful dataset or schema extensions.

4 Group Work Policy

This project must be completed in groups of two students. Individual submissions are not allowed. Each group must document each member's tasks and contributions in the final report. If effort is significantly imbalanced, grades may be adjusted.

5 Submission Summary

Phase	Due Date	Submission Requirements
Team Formation	**April 10**	Submit your team members' names and NetIDs using the released form.
Phase 1: ER Design	**April 25**	ER diagram (course notation), supporting assumptions/documentation, and one compressed file submitted on Canvas.
Phase 2: Relational Schema Design	**May 5**	One recreatable PostgreSQL `.sql` script with tables, primary keys, foreign keys, and enforceable constraints; submitted on Canvas.
Phase 3: Implementation	**June 8**	Running implementation (database + client app), source code, final report/documentation, and scheduled demo presentation. Dataset updates (if any) must be documented.
Online Demos	**Final Week**	Slots for demoing the full project will be released.

6 Requirement Analysis

This section defines the core entities, required attributes, permissions, and business constraints for the Online Auction and Bidding System.

6.1 User

Purpose: Represents any person who can access platform functionality.
Attributes:

- `phoneNum` (required)
- `login` (required)
- `password` (required)
- `role` (required)
- `address` (required)
- `favoriteCategory` (optional)

Role and profile rules:

- New accounts are assigned the default role `Buyer`.
- Only `Admin` can change a user's role to `Seller` or `Admin`.
- Users can update profile fields except `login` and `role`.

User actions after login:

- browse items;
- search auctions;
- place bids;
- view auction status;
- view/edit profile.

Additional privileges:

- `Seller`: create and manage auction listings.
- `Admin`: manage users and monitor auctions.

6.2 Item

Purpose: Represents a product listed for auction.

Attributes:

- `itemID` (required)
- `itemName` (required)
- `category` (required)
- `startingPrice` (required)
- `description` (optional)
- `condition` (optional)
- `imageURL` (optional)

Access rules:

- Sellers can create and update their own items.
- Admins can manage or remove items when needed.

6.3 Auction

Purpose: Represents the bidding process between buyers and sellers.

Attributes:

- `auctionID` (required)
- `itemID` (required)
- `sellerLogin` (required)
- `currentHighestBid` (required)
- `auctionStatus` (required)

Lifecycle rules:

- On creation, the system assigns a unique `auctionID`.
- New auctions start with status `Active`.
- The auction Seller can end the auction at any time.
- At auction end, the highest bidder becomes the winner.
- Completed auctions change status to `Closed`.

6.4 Bid

Purpose: Represents an offer submitted by a buyer on an active auction.

Attributes:

- `bidID` (required)
- `auctionID` (required)
- `buyerLogin` (required)
- `bidAmount` (required)
- `bidTimestamp` (required)

Constraints:

- Each new bid must be greater than the current highest bid.
- A seller cannot bid on their own auction.

6.5 Payment

Purpose: Represents payment completion by the winning buyer.

Attributes:

- `paymentID` (required)
- `auctionID` (required)
- `buyerLogin` (required)
- `amount` (required)
- `paymentStatus` (required)

Allowed payment status values:

- `Pending`
- `Completed`
- `Failed`

6.6 Shipment

Purpose: Represents delivery after payment completion.

Attributes:

- `shipmentID` (required)
- `auctionID` (required)

- `address` (required)
- `shipmentStatus` (required)
- `trackingNumber` (optional)

Allowed shipment status values:

- `Pending`
- `Shipped`
- `Delivered`